PERSONAL REFLECTION & AI USE DISCLOSURE FORM

BIL 476 / BIL 573 - Data Mining Course Project | TOBB University of Economics and Technology | Summer 2026

Submitted as a separate PDF with the main report.

STUDENT INFORMATION

Full Name: Remzi Mert Tehneldere

Student ID: 211401022

Chosen Topic: Topic 7 - Classification (Advanced Methods)

Project Title: Student Dropout Prediction using Data Mining Techniques

1. PROJECT REFLECTION

1. List the three most important data mining concepts or practical skills you learned:

(1) Building a complete, leakage-free preprocessing pipeline - separating categorical and numeric features, applying one-hot encoding and standardization, and fitting the scaler only on the training data. (2) Comparing multiple classifiers fairly using several complementary metrics (accuracy, F1, ROC-AUC, PR-AUC) instead of relying on accuracy alone. (3) Diagnosing overfitting through the train-test accuracy gap and validating stability with five-fold cross-validation.

2. What was the main challenge you faced, and how did you address it?

The main challenge was interpreting the fact that different models were 'best' on different metrics: Random Forest had the highest accuracy, but Logistic Regression had the highest ROC-AUC, PR-AUC and recall. I addressed this by adding a train-test gap analysis, which revealed that the tree ensembles were overfitting (training accuracy near 0.99), and by reasoning about which metric matters most for a dropout-support system (recall on the Dropout class). This turned a confusing result into a clear, defensible conclusion.

3. What was your most important finding, limitation, or lesson from the results?

The most important finding is that early academic performance - the number of approved curricular units and grades in the first two semesters - is by far the strongest predictor of dropout. The key lesson is that the 'best' model depends on the objective, not on a single headline number. The main limitation is that the data come from a single institution and that the strongest features are only available after two semesters, which limits how early a confident prediction can be made.

2. TOOLS AND REPRODUCIBILITY

Programming language and key libraries: Python; scikit-learn, XGBoost, pandas, numpy, matplotlib, seaborn.

Report writing tool: Report writing tool: LaTeX (IEEE conference template), exported to PDF.

Repository (GitHub) link: https://github.com/rmtehneldere/student-dropout-prediction

3. AI USE DISCLOSURE

Did you use any AI tools? [ ] No [X] Yes. Tool(s): Claude (Anthropic)

AI use level scale: *0 not used; 1 minimal hints; 2 low support; 3 regular support for specific sub-tasks; 4 high support for critical sub-tasks; 5 very high contribution; 6 AI-led. Levels 5-6 may indicate inappropriate use.*

Project stage	Level	Brief explanation and verification
Conceptualization and research question	1	The problem, dataset choice and research question were my own; AI was used only to sanity-check phrasing.
Literature search and synthesis	2	I selected and read the references; AI helped locate candidate sources, which I verified against the originals.
Methodology, coding, debugging, analysis	3	I designed the pipeline and decisions; AI gave regular support for coding and debugging. I ran all code myself and verified every output.
Report writing, language, formatting	3	The structure, arguments and results are mine; AI helped with language polishing and IEEE formatting. I reviewed and finalized all wording.
Interpretation, discussion, conclusion	2	The interpretation of the results and the conclusions are my own; AI was used only to improve clarity of expression.

How did you verify AI-assisted code/text/results?

I executed all code myself in VS Code and confirmed that the pipeline runs end-to-end without errors and reproduces identical results on repeated runs (fixed random seed). I checked every reported metric against the program's own console output and the saved CSV tables, and I confirmed the confusion-matrix counts by hand. All references were verified against their original publications, and I made sure I can explain every design decision and every line of the submitted code.

4. COURSE FEEDBACK

Project difficulty: [ ] Too easy [X] Appropriate [ ] Too hard

Project duration: [ ] Too short [X] Appropriate [ ] Too long

One suggestion to improve this project:

Providing a short optional checklist that maps each required report section to the grading rubric would help students make sure every graded component is covered before submission.

ACADEMIC INTEGRITY STATEMENT

I declare that this project is my own individual work; all sources, tools, datasets, and AI assistance have been properly cited or declared; results are genuine and not fabricated; and I understand and can explain the submitted report and code.

Student: Remzi Mert Tehneldere

Date: 31 / 07 / 2026